

Hao Wu

Ph.D. Student | Research Intern @ Tencent IEG, LightSpeed Studios

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I am a Ph.D. student at **Tsinghua University** and currently a Research Intern at **Tencent IEG, LightSpeed Studios** working on video generation. Previously, I conducted research at **Tencent CSIG, Tencent Jarvis Lab**; I graduated from the Department of Computer Science at the **University of Science and Technology of China (USTC)**, during which I was also a joint training student in the large model training group of the Machine Learning Platform Department at **Tencent Hunyuan**.

Research Interests: My research lies at the intersection of (1) **Robot Video Generation & World Models**; (2) **Multimodal Large Language Models (VLMs)**; (3) **Agentic Reasoning Systems**; and (4) **Large-scale Spatiotemporal Forecasting**. Broadly, I aim to build intelligent systems that can *understand, predict, and reason about* the physical world across space, time, and modalities. My work has appeared in top venues including **ICLR, NeurIPS, ICML, KDD, AAAI, ICCV, ACM MM, TKDE, and TPAMI**, with **1,200+ Google Scholar citations** and nearly **30 CCF-A publications** (including ~20 as first or corresponding author). I serve as a reviewer for **ICLR, NeurIPS, ICML, KDD, ICCV, AAAI, ACM MM, TPAMI, and TKDE**.

RESEARCH EXPERIENCE

Research Intern, Tencent IEG, LightSpeed Studios	May 2026 – Present
• Researching video generation world models , with a focus on long-horizon video generation, controllable dynamics, and physically grounded simulation for interactive scenarios.	
Research Intern, Tencent CSIG, Tencent Jarvis Lab	Aug 2025 – May 2026
• Continued research on multimodal foundation models, agent systems, and world models in industrial-scale settings.	
Research Intern, Tencent Hunyuan, Machine Learning Platform Department	Aug 2023 – Jul. 2025
• Worked on large models, world models, and multimodal generative modeling in Tencent Hunyuan.	
Online Research Intern, UCLA	May 2023 – May 2024
• Advisor: Xiao Luo . Remote research on multimodal learning and dynamics modeling.	
Research Intern, CityMind Lab, HKUST (Guangzhou)	Mar 2023 – Aug 2023
• Advisors: Yuxuan Liang and Kun Wang . Early-stage research on spatiotemporal modeling.	

SELECTED AWARDS & HONORS

Outstanding Graduate, USTC × Tencent Joint Training Program	2025
National Scholarship (Top 1% in China), USTC	2022
First-Class Scholarship, USTC	2022

PUBLICATIONS

† equal / co-first author, ∧ co-first author, # / * corresponding author. 61 works listed (incl. preprints & under-review); nearly 30 CCF-A publications, ~20 as first or corresponding author.

Robot Video Generation & World Models

- RoboAlign-R1: Distilled Multimodal Reward Alignment for Robot Video World Models**
H. Wu, Y. Li, Y. Gao, F. Xu, F. Zhang, K. Wang, P. Zhao, Q. Wang, Y. Zhao, W. Wang, Y. Tian, X. Wu, X. Huang# · **Preprint (arXiv:2605.03821), 2026**
[Project Page]
- Spatiotemporal Forecasting as Planning: A Model-Based Reinforcement Learning Approach with Generative World Models**
H. Wu, Y. Gao, X. Shi, S. Li, F. Xu, F. Zhang, Z. Zhu, W. Wang, X. Luo, K. Wang · **Preprint (arXiv:2510.04020), 2025**

Multimodal LLMs, Agents & Trustworthy AI

3. **Latent Visual Cache for Video Reasoning**
Y. Zhang, Z. Xu, **H. Wu**, Y. Li, D. Yin, X. Sun, P.S. Yu · [Preprint \(arXiv:2607.02607\)](#), 2026
4. **Orchestra-or: Omnimodal Agent Orchestration**
F. Zhang, V. Zhang, S. Qian, H. Li, **H. Wu**, J. Wu, D. Zhou, Z. Zhu, Z. Lian · [Preprint \(arXiv:2606.13707\)](#), 2026
5. **From Chatbot to Digital Colleague: The Paradigm Shift Toward Persistent Autonomous AI**
Y. Zhang, Z. Liu, J. Zhu, S. Wang, X. Chen, H. Huang, J. Kuang, S. Chen, **H. Wu** · [Preprint \(arXiv:2606.14502\)](#), 2026
6. **S³-MSD: Large Vision-Language Model for Explainable and Generalizable Multi-modal Sarcasm Detection**
Z. Zhu, F. Zhang, Y. Zhang, J. Sun, G. Hu, **H. Wu**, Y. Chen, B. Xing, X. Wu · [AAAI](#), 2026
7. **From Instruction Following to Cognitive Navigation: A Survey on the Evolution of Vision-and-Language Navigation**
K. Lyu, K. Wu, P. Li, W. Song, D. Wu, J. He, J. Chen, N. Yang, Z. Han, K. Luo, **H. Wu** · [Preprint](#), 2026
8. **Struct-Searcher: Agentic Structural Thinking Advances Multimodal Deep Information Seeking**
F. Zhang, V. Zhang, S. Qian, H. Li, Z. Lian, **H. Wu**, Y. Gao, X. Geng, X. Wang · [Preprint \(arXiv:2606.07689\)](#), 2026
9. **ToolSelf: Unifying Task Execution and Self-Reconfiguration via Tool-Driven Emergent Adaptation**
J. Zhou, S. Wang, D. Deng, J. Lu, J. Su, Q. Li, J. Gao, **H. Wu**, J. Jiang, L. Kong · [Preprint \(arXiv:2602.07883\)](#), 2026
10. **OMNIFLOW: A Physics-Grounded Multimodal Agent for Generalized Scientific Reasoning**
H. Wu, Y. Zhang, Y. Gao, F. Xu, F. Zhang, R. Xie, R. Gou, Y. Liang, X. Huang · [Preprint \(arXiv:2603.15797\)](#), 2026
11. **SEAT: Calibrating Inference Time Alignment with Sequence-level Risk Accumulation**
S. Tan†, Z. Dong†, W. Ju#, Y. Fu#, **H. Wu**, K. Wang, Y. Wang, Z. Qiao · [ACL](#), 2026
12. **MME-Emotion: A Holistic Evaluation Benchmark for Emotional Intelligence in Multimodal Large Language Models**
F. Zhang, Z. Cheng, C. Deng, H. Li, Z. Lian, Q. Chen, H. Liu, W. Wang, **H. Wu**, H. Wang, Y. Zheng, X. Peng, X. Wu, K. Wang, X. Li, J. Ye, P.-A. Heng · [ICLR](#), 2026
13. **SaFeR-VLM: Toward Safety-aware Fine-grained Reasoning in Multimodal Models**
H. Yi, K. Wang, Q. Li, M. Yu, L. Lin, G. Xi, **H. Wu**#, X. Hu, K. Li, Y. Liu · [IEEE TPAMI](#), Under Review (arXiv:2510.06871), 2025
14. **A Comprehensive Survey in LLM(-Agent) Full Stack Safety: Data, Training and Deployment**
K. Wang, G. Zhang, Z. Zhou, J. Wu, M. Yu, S. Zhao, C. Yin, J. Fu, Y. Yan, H. Luo, **H. Wu** · [Preprint \(arXiv:2504.15585\)](#), 2025
15. **Probing Scientific General Intelligence of LLMs with Scientist-Aligned Workflows**
W. Xu, Y. Zhou, Y. Zhou, Q. Cao, S. Li, J. Bu, B. Liu, Y. Chen, X. He, X. Zhao, **H. Wu** · [Preprint \(arXiv:2512.16969\)](#), 2025
16. **Rethinking Facial Expression Recognition in the Era of Multimodal Large Language Models: Benchmark, Datasets, and Beyond**
F. Zhang, H. Li, S. Qian, X. Wang, Z. Lian, **H. Wu**, Z. Zhu, Y. Gao, Q. Li, Y. Zheng · [Preprint \(arXiv:2511.00389\)](#), 2025
17. **CellVerse: Do Large Language Models Really Understand Cell Biology?**
F. Zhang, T. Liu, Z. Zhu, **H. Wu**, H. Wang, D. Zhou, Y. Zheng, K. Wang, X. Wu · [NeurIPS](#), 2025

Large-scale Spatiotemporal & Earth-System Forecasting

18. **Tyche: One Step Flow for Efficient Probabilistic Weather Forecasting**
F. Xu, Y. Gao, K. Wang, R. Su, F. Ling, **H. Wu**, W. Ouyang · [Preprint \(arXiv:2605.06916\)](#), 2026
19. **Physics-Constrained Network for Enhanced Extended-Range Precipitation Forecasting in East Asia**
Y. Wang, H. Chen, **H. Wu**, J. Liu, H. Yuan, S. Cao, T. Wang, B. Zhuang · [Geophysical Research Letters](#), 2026
20. **PnP-Corrector: A Universal Correction Framework for Coupled Spatiotemporal Forecasting**
H. Wu, F. Xu, Y. Lu, P. Zhao, F. Zhang, H. Jia, Y. Liang, R. Gou, Q. Wen, X. Wu, X. Huang, Y. Gao · [ICML](#), 2026
21. **NuwaDynamics+: A Causality-Aware Generative Framework for Spatio-Temporal Representation Learning**
K. Wang, Y. Duan, **H. Wu**, J. Zhao, K. Wang, Z. Zhou, Y. Liang, X. Wang*, Y. Wang*, Y. Zheng, X. Li · [IEEE TPAMI](#), 2026
22. **NeuralOM: Neural Ocean Model for Subseasonal-to-Seasonal Simulation**
Y. Gao†, **H. Wu**†, F. Xu, Y. Xiang, R. Gou, R. Shu, Q. Wen, X. Wu, K. Wang*, X. Huang* · [AAAI](#), 2026
23. **TritonCast: Advanced Long-term Earth System Forecasting**
H. Wu, Y. Gao, R. Gou, X. Wu, C. Wu, H. Yi, J. Brandstetter, F. Xu, N. Boers, K. Wang, X. Huang · [Nature Machine Intelligence](#), Under Review (arXiv:2505.19432), 2025
24. **Turb-Lr: Achieving Long-term Turbulence Tracing by Tackling Spectral Bias**

- H. Wu**, Y. Gao, C. Liu, F. Xu, F. Zhang, Z. Zhu, Y. Li, X. Wu, Y. Liang, L. Liu · [Preprint \(arXiv:2505.19038\)](#), 2025
25. **OneForecast: A Universal Framework for Global and Regional Weather Forecasting**
Y. Gao, **H. Wu**, R. Shu, H. Dong, F. Xu, R. Chen, Y. Yan, Q. Wen, X. Hu, K. Wang, J. Wu, L. Qing, H. Xiong, X. Huang# · [ICML, 2025](#)
 26. **Frequency-Aligned Knowledge Distillation for Lightweight Spatiotemporal Forecasting**
Y. Li, C. Yang, H. Zeng, Z. Dong, Z. An, Y. Xu, Y. Tian, **H. Wu**# · [ICCV, 2025](#)
 27. **DynST: Dynamic Sparse Training for Resource-Constrained Spatio-Temporal Forecasting**
H. Wu, H. Wen, G. Zhang, Y. Xia, K. Wang, Y. Liang, Y. Zheng, K. Wang# · [KDD, 2025](#)
 28. **S3-Fire: Enhanced Generalization in Spatiotemporal Fire Prediction with Selective State and Spectral Spaces**
H. Wu, J. Ji · [Expert Systems with Applications, 2025](#)
 29. **Ocean-E2E: Hybrid Physics-based and Data-driven Global Forecasting of Extreme Marine Heatwaves with End-to-End Neural Assimilation**
R. Shu, Y. Gao, **H. Wu**, R. Gou, K. Wang, Y. Xiang, F. Xu, Q. Wen, X. Huang · [Preprint \(arXiv:2505.22071\)](#), 2025
 30. **Learning from History: A Retrieval-Augmented Framework for Spatiotemporal Prediction**
H. Jia, P. Zhao, **H. Wu**, Y. Gao, Y. Tao, B. Cui · [Preprint \(arXiv:2510.24049\)](#), 2025
 31. **Advancing Ocean State Estimation with Efficient and Scalable AI**
Y. Xiang, Y. Gao, **H. Wu**, Q. Zhang, R. Shu, X. Zhou, X. Wu, X. Huang · [Preprint \(arXiv:2511.06041\)](#), 2025
 32. **Unlocking Out-of-Distribution Generalization in Dynamics through Physics-Guided Augmentation**
F. Xu, **H. Wu**, K. Wang, N. Wang, Q. Wen, X. Wu, W. Gong, X. Zhao · [Preprint \(arXiv:2510.24216\)](#), 2025
 33. **VISION: Prompting Ocean Vertical Velocity Reconstruction from Incomplete Observations**
Y. Gao, **H. Wu**, Q. Wen, K. Wang, X. Wu, X. Huang · [Preprint \(arXiv:2509.21477\)](#), 2025
 34. **Spatio-temporal Fluid Dynamics Modeling via Physical-Awareness and Parameter Diffusion Guidance**
H. Wu, F. Xu, Y. Duan, Z. Niu, W. Wang, G. Lu, K. Wang, Y. Liang, Y. Wang · [Preprint \(arXiv:2403.13850\)](#), 2024
 35. **EarthFarseer: Versatile Spatio-Temporal Dynamical Systems Modeling in One Model**
H. Wu, Y. Liang, W. Xiong#, Z. Zhou, W. Huang, S. Wang, K. Wang# · [AAAI, 2024](#)
 36. **NuwaDynamics: Discovering and Updating in Causal Spatio-Temporal Modeling**
K. Wang^, **H. Wu**^, Y. Duan, G. Zhang, K. Wang, X. Peng, Y. Zheng, Y. Liang#, Y. Wang# · [ICLR \(Spotlight\), 2024](#)
 37. **Modeling Spatio-Temporal Dynamical Systems with Neural Discrete Learning and Levels-of-Experts**
K. Wang^, **H. Wu**^, G. Zhang, J. Fang, Y. Liang, Y. Wu, R. Zimmermann, Y. Wang# · [IEEE TKDE, 2024](#)
 38. **PastNet: Introducing Physical Inductive Biases for Spatio-Temporal Video Prediction**
H. Wu, W. Xiong, F. Xu, X. Luo#, C. Chen, X.-S. Hua, H. Wang# · [ACM MM, 2024](#)
 39. **Causal Deciphering and Inpainting in Spatio-Temporal Dynamics via Diffusion Model**
Y. Duan, J. Zhao, J. Mao, **H. Wu**, J. Xu, C. Ma, K. Wang, K. Wang, X. Li · [NeurIPS, 2024](#)
 40. **AI-GOMS: Large AI-Driven Global Ocean Modeling System**
W. Xiong, Y. Xiang, **H. Wu**, S. Zhou, Y. Sun, M. Ma, X. Huang · [Preprint \(arXiv:2308.03152\)](#), 2023
 41. **Spatio-temporal Twins with a Cache for Modeling Long-term System Dynamics**
H. Wu, K. Wang, F. Xu, Y. Li, X. Wang, W. Wang, H. Wang, X. Luo · [Preprint, 2023](#)
 42. **Adaptive Universal Network for Ocean Vertical Velocity Reconstruction**
Y. Gao, W. Xiong, **H. Wu**, R. Shu, X. Huang · [AGU Fall Meeting, 2024](#)
 43. **A Scale-Aware Framework for Regional to Global Marine Heatwaves Forecast**
R. Shu, W. Xiong, **H. Wu**, Y. Gao, R. Gou, X. Huang · [AGU Fall Meeting, 2024](#)

Scientific Machine Learning & Neural Operators

- 44. **Differential-Integral Neural Operator for Long-Term Turbulence Forecasting**
H. Wu, Y. Gao, F. Xu, F. Zhang, Q. Wen, K. Wang, X. Huang, X. Wu · **KDD, 2026**
- 45. **Advanced Global Wildfire Activity Modeling with Hierarchical Graph ODE**
F. Xu, W. Gong, H. Wu, L. Peng, N. Wang, Q. Wen, X. Wu, K. Wang, X. Zhao · **KDD, 2026**
- 46. **An Exterior-Embedding Neural Operator Framework for Preserving Conservation Laws**
H. Dong, H. Wang, H. Wu, Z. Zhuang, X. Yang, R. Shu, Y. Gao, X. Huang · **KDD, 2026**
- 47. **Evolving Multimodal Models for Physical Dynamics: A Multi-objective Neuroevolution Approach**
Y. Li, J. Dong, J. Liu, P. Koniusz, H. Zeng, C. Yang, J. Liu, Y. Tian, T. Huang, H. Wu · **IEEE Trans. Evolutionary Computation, 2026**
- 48. **Breaking the Discretization Barrier of Continuous Physics Simulation Learning**
F. Xu, H. Wu, N. Wang, L. Peng, K. Wang, W. Gong, X. Zhao · **NeurIPS, 2025**
- 49. **Open-CK: A Large Multi-Physics Fields Coupling Benchmarks in Combustion Kinetics**
Z. Fei, F. Xu, J. Mao, Y. Liang, Q. Wen, K. Wang, H. Wu#, Y. Wang · **ICLR, 2025**
- 50. **FourierFlow: Frequency-aware Flow Matching for Generative Turbulence Modeling**
H. Wang, J. Pan, H. Wu, F. Zhang, T. Wu · **Preprint (arXiv:2506.00862), 2025**
- 51. **PURE: Prompt Evolution with Graph ODE for Out-of-distribution Fluid Dynamics Modeling**
H. Wu, C. Wang, F. Xu, J. Xue, C. Chen, X.-S. Hua, X. Luo# · **NeurIPS, 2024**
- 52. **Prometheus: Out-of-distribution Fluid Dynamics Modeling with Disentangled Graph ODE**
H. Wu, H. Wang, K. Wang, W. Wang, Y. Tao, C. Chen, X.-S. Hua, X. Luo# · **ICML, 2024**
- 53. **Neural Manifold Operators for Learning the Evolution of Physical Dynamics**
H. Wu, K. Weng, S. Zhou, X. Huang#, W. Xiong# · **KDD, 2024**
- 54. **IDEA: An Invariant Perspective for Efficient Domain Adaptive Image Retrieval**
H. Wang^, H. Wu^, J. Sun, S. Zhang, C. Chen, X.-S. Hua, X. Luo# · **NeurIPS, 2023**

Graph Learning, Optimization & Other Applications

- 55. **Out-of-Distribution-Resistant Evaluations for Explanations of Graph Neural Networks**
J. Fang, H. Wu, A. Zhang, T. Chen, K. Wang, Y. Liang, R. Zimmermann · **IEEE TPAMI, 2026**
- 56. **Revisiting Graph-based Fraud Detection in Sight of Heterophily and Spectrum**
F. Xu, N. Wang, H. Wu, X. Wen, X. Zhao, H. Wan · **AAAI, 2024**
- 57. **Surge Phenomenon in Optimal Learning Rate and Batch Size Scaling**
S. Li, P. Zhao, H. Zhang, X. Sun, H. Wu, D. Jiao, W. Wang, C. Liu, Z. Fang · **NeurIPS, 2024**
- 58. **Cat-GNN: Enhancing Credit Card Fraud Detection via Causal Temporal Graph Neural Networks**
Y. Duan, G. Zhang, S. Wang, X. Peng, Z. Wang, J. Mao, H. Wu, X. Jiang, K. Wang · **Preprint (arXiv:2402.14708), 2024**
- 59. **Gladformer: A Mixed Perspective for Graph-level Anomaly Detection**
F. Xu, N. Wang, H. Wu, X. Wen, D. Zhang, S. Lu, B. Li, W. Gong, H. Wan · **ECML-PKDD, 2024**
- 60. **ForecastGrapher: Redefining Multivariate Time Series Forecasting with Graph Neural Networks**
W. Cai, K. Wang, H. Wu, X. Chen, Y. Wu · **Preprint (arXiv:2405.18036), 2024**
- 61. **SLFNet: Generating Semantic Logic Forms from Natural Language using Semantic Probability Graphs**
H. Wu, F. Xu · **Preprint (arXiv:2403.19936), 2024**

ACADEMIC SERVICE

Reviewer: ICLR, NeurIPS, ICML, KDD, ICCV, AAAI, ACM MM, TPAMI, TKDE

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java

Frameworks & Tools: PyTorch, JAX, TensorFlow, DeepSpeed, Megatron, Git, Linux, L^AT_EX

Research Areas: Robot Video Generation & World Models, Multimodal LLMs / VLMs, Agent Systems, Reinforcement Learning from Human / AI Feedback, Spatiotemporal & Earth-System Forecasting, Neural Operators